

Enhanced Anomaly Detection with Hybrid Siamese Network

Objective:

The goal of this project was to improve anomaly detection in time-series data using an advanced Hybrid Siamese Network. The model was trained and evaluated on the NAB (Numenta Anomaly Benchmark) dataset, specifically using the `speed_6005.csv` file.

Data Preprocessing:

1. Loading and Cleaning Data:
 - The data was loaded from the `csv_path`, and the `timestamp` column was converted to the `datetime` format.
 - Missing values in the `value` column were handled using time-based interpolation and then scaled using `StandardScaler`.
 2. Anomaly Labeling:
 - Anomalies were identified using labels from the `combined_labels.json` file, with one anomaly identified at index 2386.
 3. Sliding Window Approach:
 - A sliding window approach was applied to the time-series data, creating 100-point windows with a step size of 5 to reduce overlap and prevent overfitting.
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Model Development:

1. Masked Autoencoder:
 - A Masked Autoencoder was used for feature extraction with convolutional layers.
 - Data augmentation, including Gaussian noise and random scaling, was applied to stabilize training.
 - The model was trained for 40 epochs, with early stopping to prevent overfitting.
 2. Siamese Network:
 - The pre-trained autoencoder's encoder was used to create embeddings for the data.
 - Hybrid triplets (anchor, positive, and negative) were generated using reconstruction errors and random normal samples.
 3. Training:
 - The Siamese network was trained for 20 epochs with triplet loss, reducing loss over time. The best model was saved using `ModelCheckpoint`.
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Anomaly Detection:

- Prototype Anomaly Scoring:
 - After training, embeddings for the train, validation, and test datasets were computed.

- Anomaly scores were calculated as the Euclidean distance between the embeddings and the normal class prototype (mean of normal embeddings).
 - The threshold for anomaly detection was set at the 90th percentile of the validation anomaly scores.
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Evaluation:

1. Metrics:

- Precision: 0.6818
- Recall: 0.7500
- F1 Score: 0.7143

Confusion Matrix:

```
[[46  7]
 [ 5 15]]
```

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- The model performed well, with balanced precision and recall, achieving an F1 score of 0.7143.

2. Threshold:

- The threshold for anomaly detection was set at 0.5953, derived from the 90th percentile of normal validation scores.
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Visualization:

A plot of the anomaly scores for the test dataset was generated, with the threshold line clearly marking the cutoff for anomaly detection.

Conclusion:

The Hybrid Siamese Network demonstrated significant improvement in detecting anomalies with an F1 score of 0.7143. The model effectively identifies anomalies, with balanced precision and recall, indicating good performance in anomaly detection tasks.

Next Steps:

1. Model Tuning: Further refinement of hyperparameters and network architecture is recommended to enhance performance.
2. Threshold Adjustment: Experimenting with other thresholds and performance metrics could improve evaluation.
3. Data Augmentation: Expanding the dataset with more anomalies or synthetic data could increase model robustness.